DIGITAL IMAGE PROCESSING (DIP)

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Lecture 1: Introduction

Fall 1405
Fall 2026

Course Outlier

In all parts, we will use the definitions in MATLAB.
Students can also use Python.

- Introduction of digital images;
- Color spaces;
- Image Enhancement;
- Image Restoration;
- Midterm exam
- Image Transforms, Eigenimages;
- Image Compression;
- Video Definition.

References:

- R. C. Gonzalez, R. E. Woods, "Digital Image Processing", 4nd edition, Prentice-Hall, 2018.
- M. Sonka, V. Hlavac and R. Boyle, "Image Processing.Analysis, and Machin Vision", 4th ed. CENGAGE Learning, 2015.
- S. E. Umbaugh, "Computer Vision and Image Analysis", 4th ed. CRC Press, 2023.
- W. K. Pratt, "Digital Image Processing", Wiley Interscience, 2nd edition, 2001.
- Al Bovik, "The Essential Guide to Image Processing", Academic Press, 2009.

Software-centric books:

- R. C. Gonzalez, R. E. Woods, S. L. Eddins, "Digital Image Processing using Matlab", 2nd edition, Pearson-Prentice-Hall, 2009, CA.
- S. Dey, "Hands-On Image Processing with Python", Packt Publisheing, 2020. (Newer edition will be in 2025.)

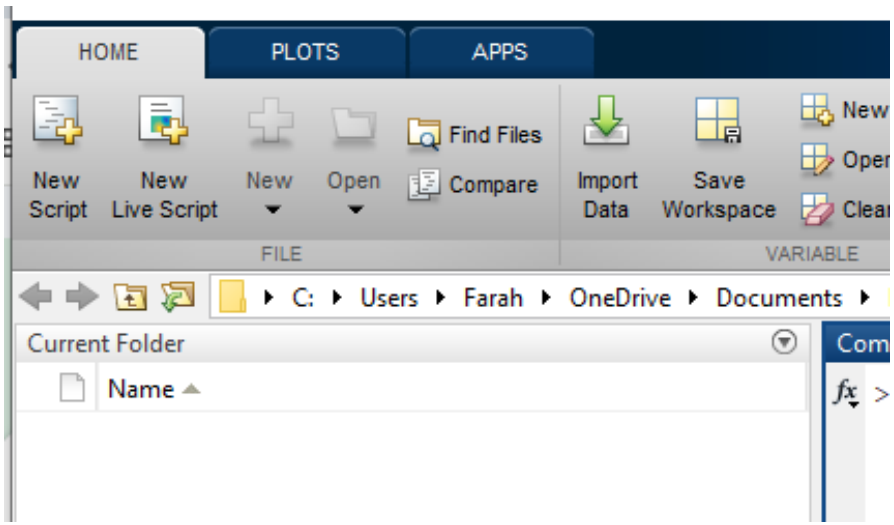
References for Slides:

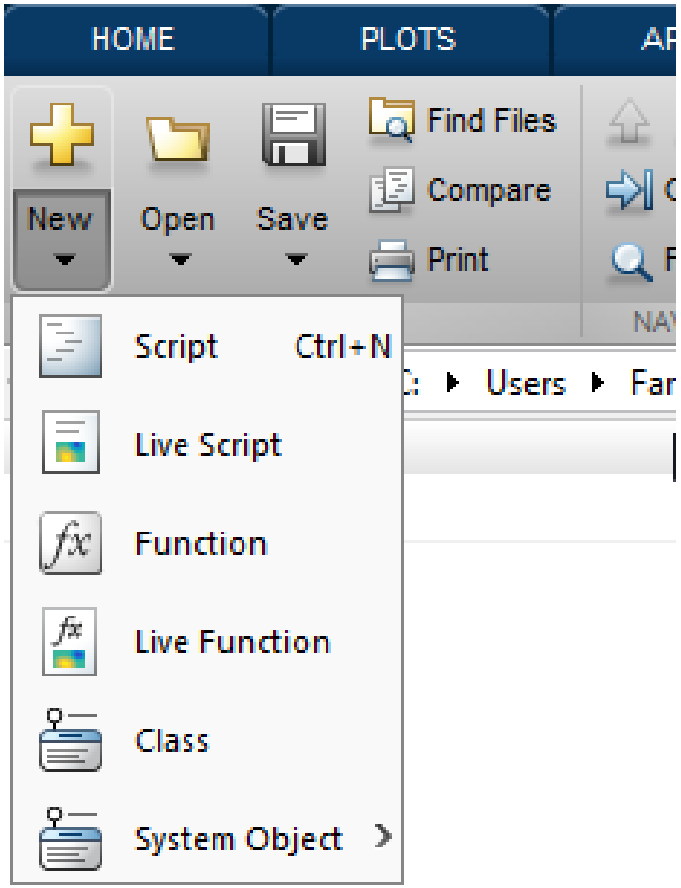
- Prof. Bernd Girod: Stanford University, USA.
- Prof. Shih – Fu Chang, Columbia University, USA.
- Prof. Sinisa Todorovic , Oregon State University, USA.
- Prof. G. Guleryuz, Polytechnic University, Brooklyn, NY , USA.
- Prof. Min Wu, Univ. of Maryland, College Park.
- Prof. Deepa Kundur, Univ. of Toronto, Canada.
- Prof. S.Ali Amirshahi, Norwegian University of Science and Technology (NTNU), Norwey.
- Prof. Zhu Li, University of Missouri, USA.
- Prof. Andrew Zisserman, Computer Vision Engineering, Oxford, UK.
- Prof. Aline.Roumy,httpInria University, France.
- Prof. Kai Zheng, University of Electronic Science and Technology, China.
- Prof. A.K. Katsaggelos, University of Northwestern, USA.

Grades

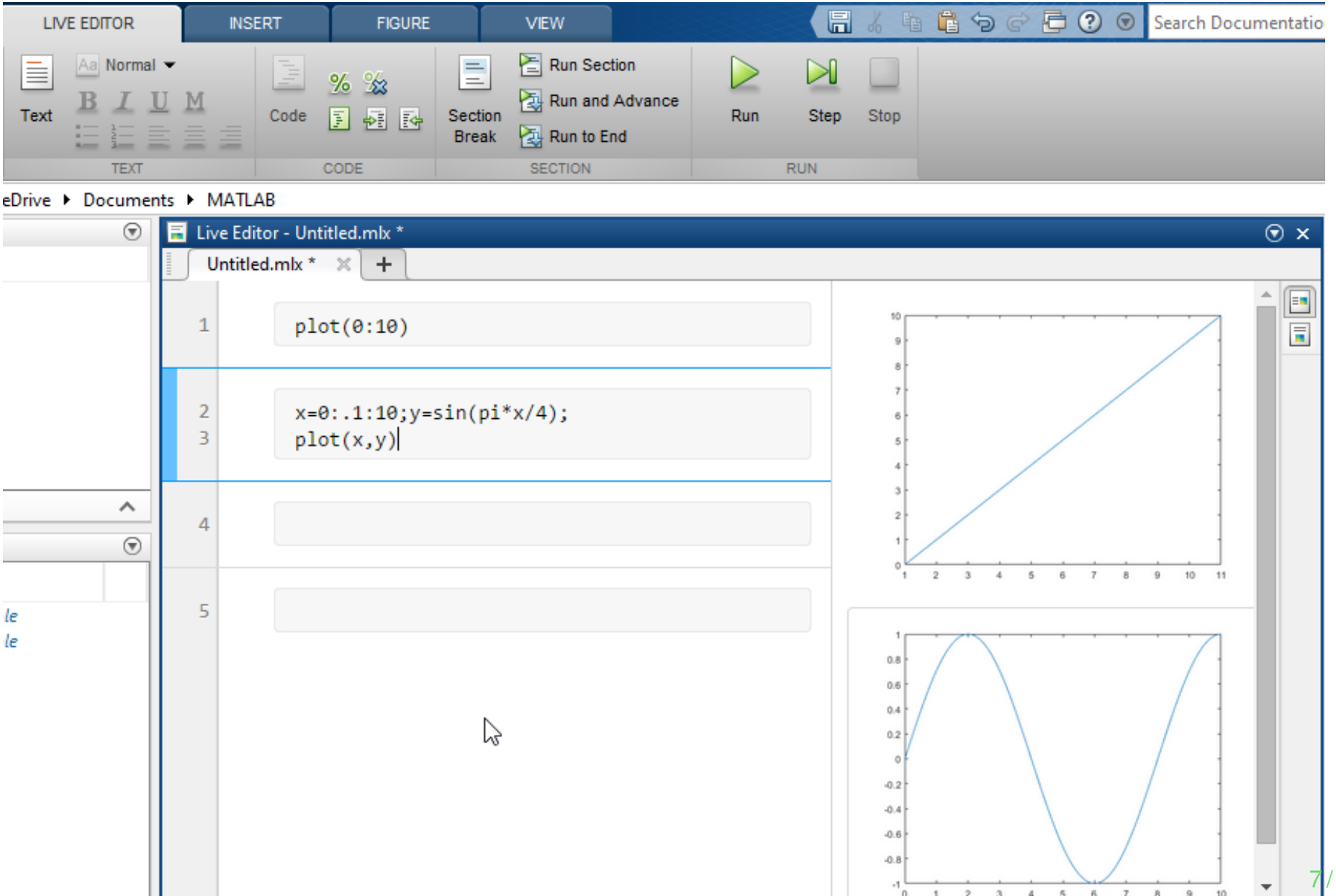
Projects (Computer HWs.) (4-5): 30%
Exercises are delivered in person during non-class hours.
Midterm Exam: 35 %
Final Exam: 35 %
Class Activation: 10%

How to Prepare HWs Reports in Matlab:

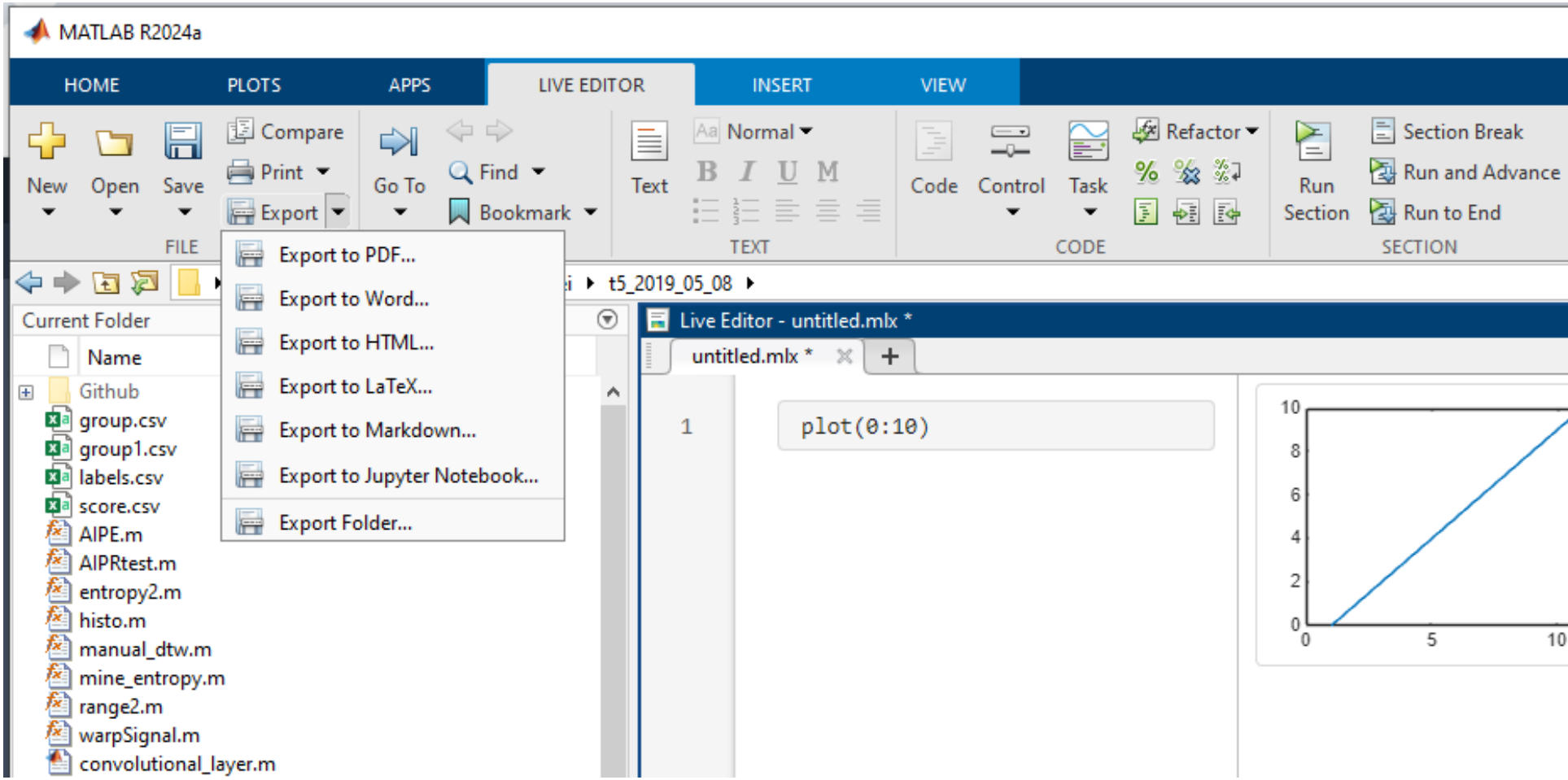




How to Prepare HWs Reports in Matlab:



How to Prepare HWs Reports in Matlab:



How to Prepare HWs Reports in Matlab:

Live Editor - untitled2.mlx *

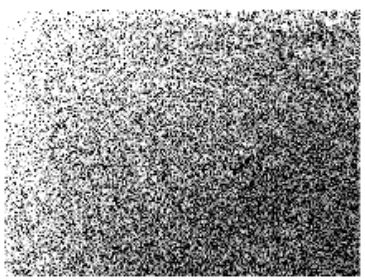
untitled2.mlx *

```
1  a=imread('car_1.jpg');
2
3  imshow(a)
```

```
4  b=double(rgb2gray(a));
5  imshow(b/255)
6
7  B=dct2(b);
8  imshow(log10(abs(B)+.001))
9
10
```







How to Prepare HWs Reports in Matlab:

```
a=imread('car_1.jpg');¶
¶
imshow(a)¶
```



¶

```
b=double(rgb2gray(a));¶
imshow(b/255)¶
```



¶

```
¶
B=dct2(b);¶
imshow(log10(abs(B)+.001))¶
```



¶

In Word, you can include an explanation of your work, reasons, and results for every required line.

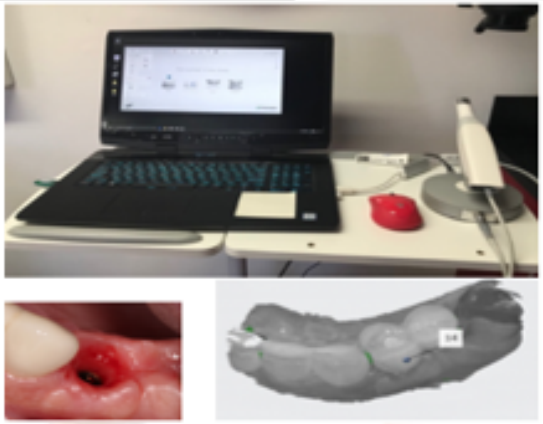
Why Image Processing?

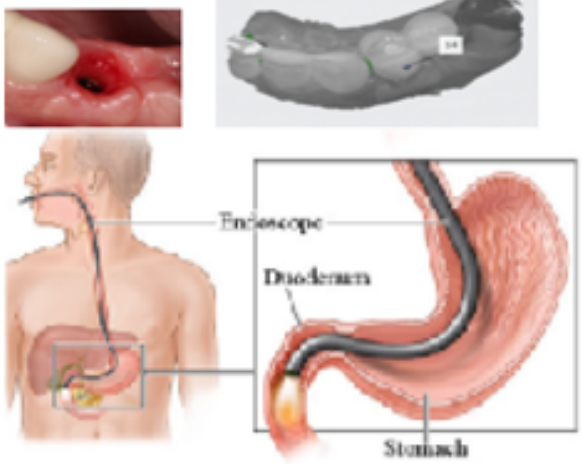
- Acquire an image
- Prepare for displaying or printing
- Enhancement and Restoration images
- Facilitate picture storage and transmission
- Extract information
- Medical Imaging



Why Image Processing? Continued:







Register from 33 source image



Iris Recognition:

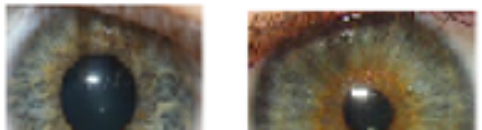


Image Processing and related Fields

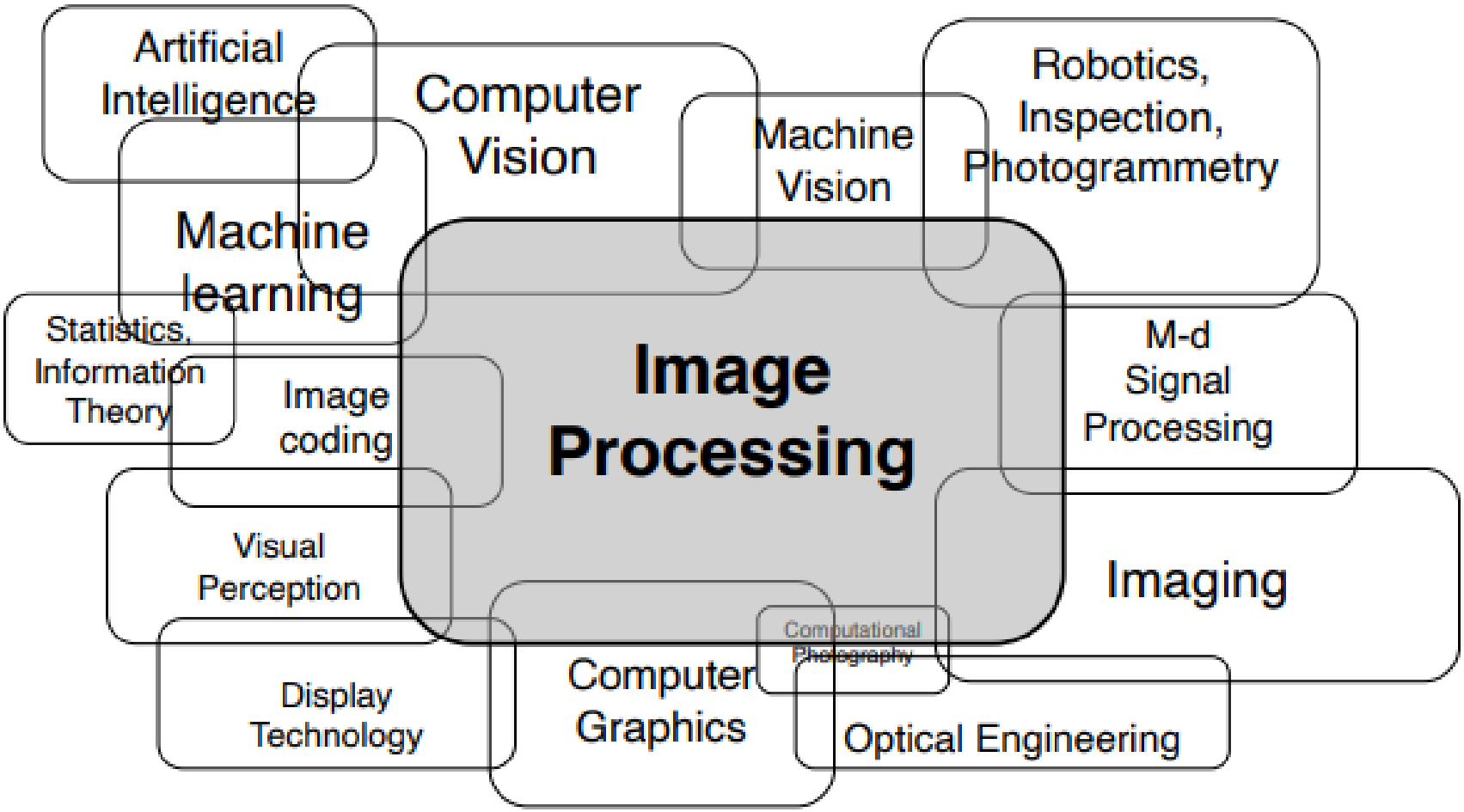
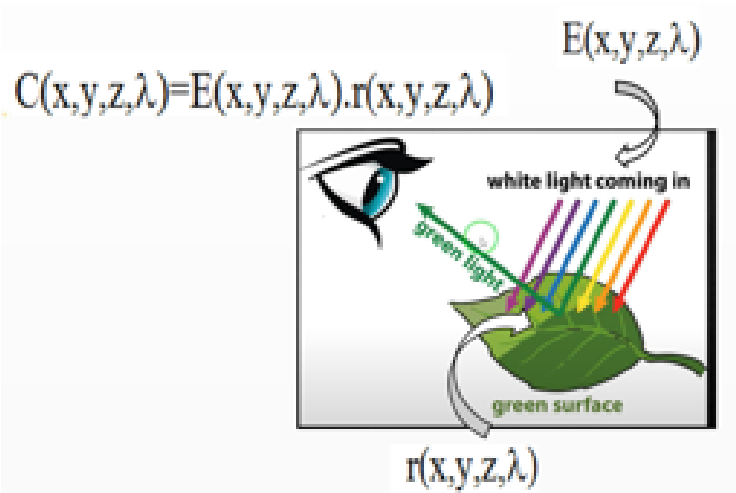


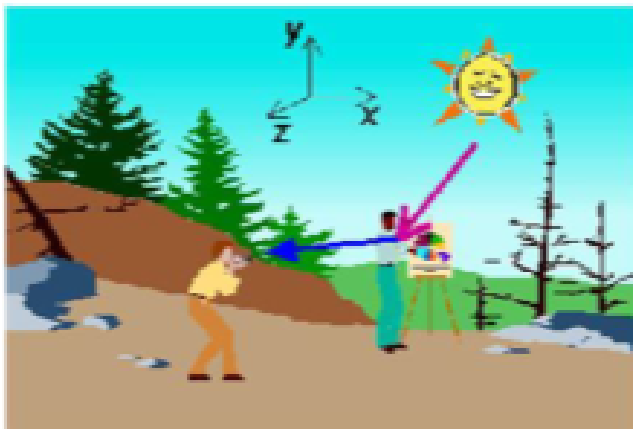
Image Acquisition; Physical Scene

An image is a physical measurement that becomes a 2-D digital signal.

Human Observation



Camera Capturing



$\text{Camera}[C(x,y,z,\lambda)] \Longrightarrow C_p(x',y',\lambda)$

$E(x, y, z, \lambda)$ is the incident light on a point (x, y, z) and wavelength λ of the source.

$r(x, y, z, \lambda)$ is the reflectivity function.

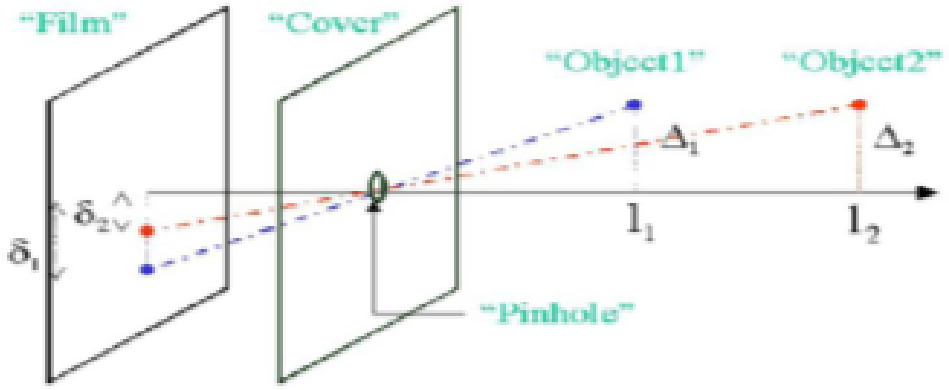
$C(x, y, z, \lambda)$ is the received light.

$C_p(x', y', \lambda)$ is the brightness on the image's point (x', y') comes from

Projections

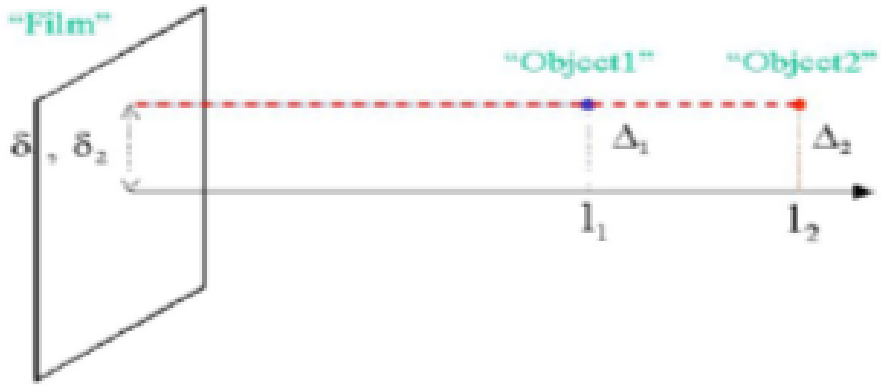
Perspective Projection

$\Delta_1 = \Delta_2,\ l_1 < l_2 \Rightarrow \delta_1 > \delta_2$



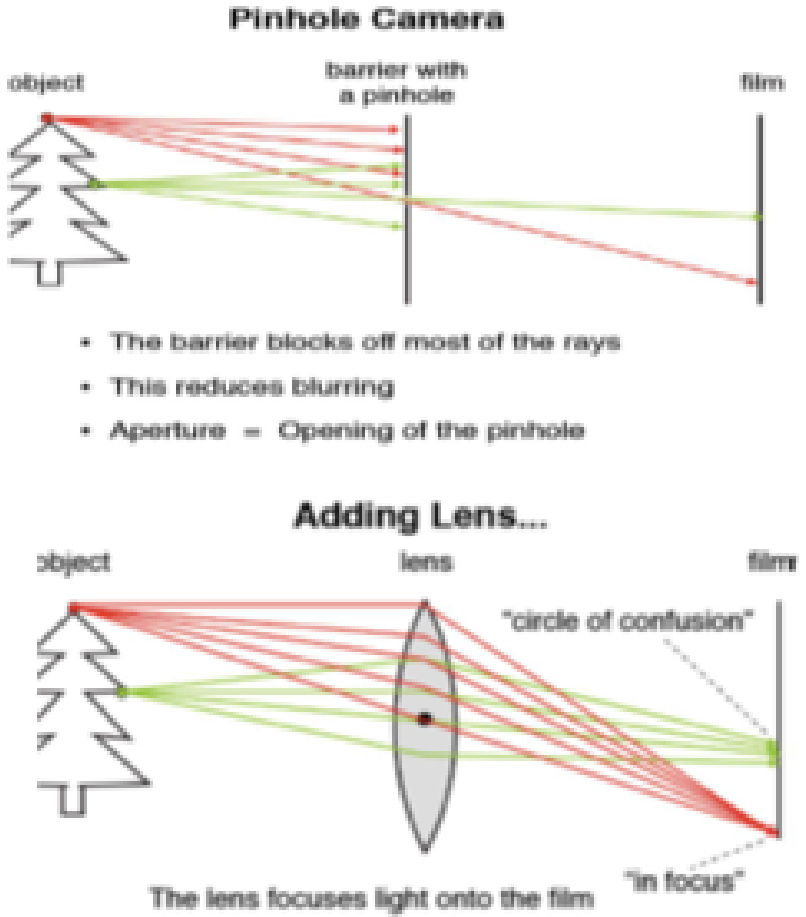
Orthogonal Projection

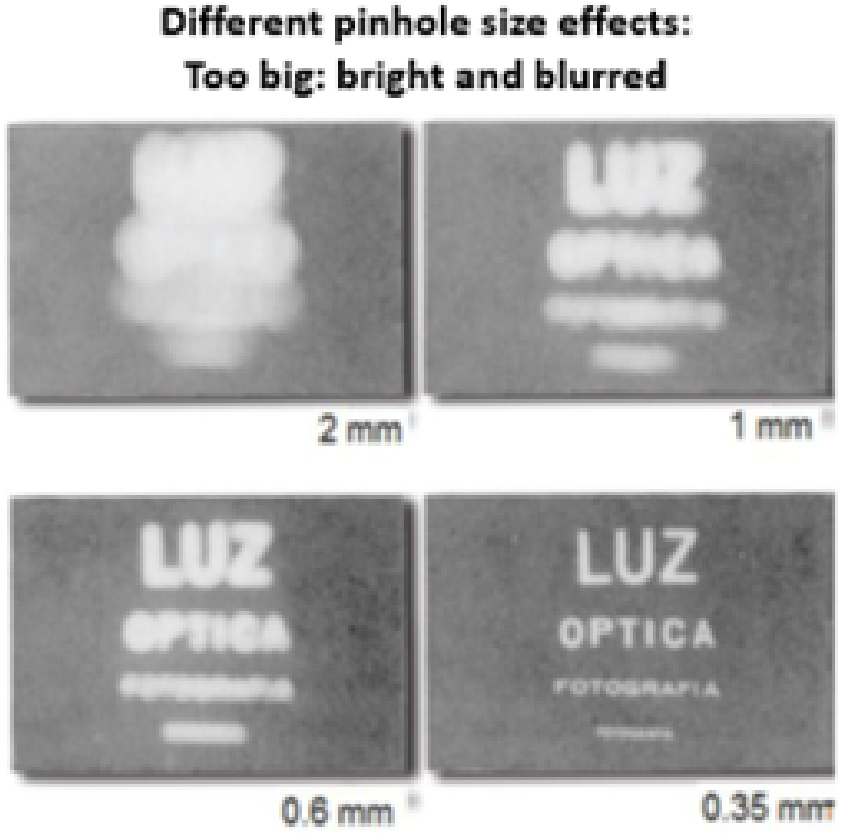
$\Delta_1 = \Delta_2,\ l_1 < l_2 \Rightarrow \delta_1 = \delta_2$



In later, in this class, the perspective projection is considered.

Pinhole Effects





Analog VS. Digital Signals

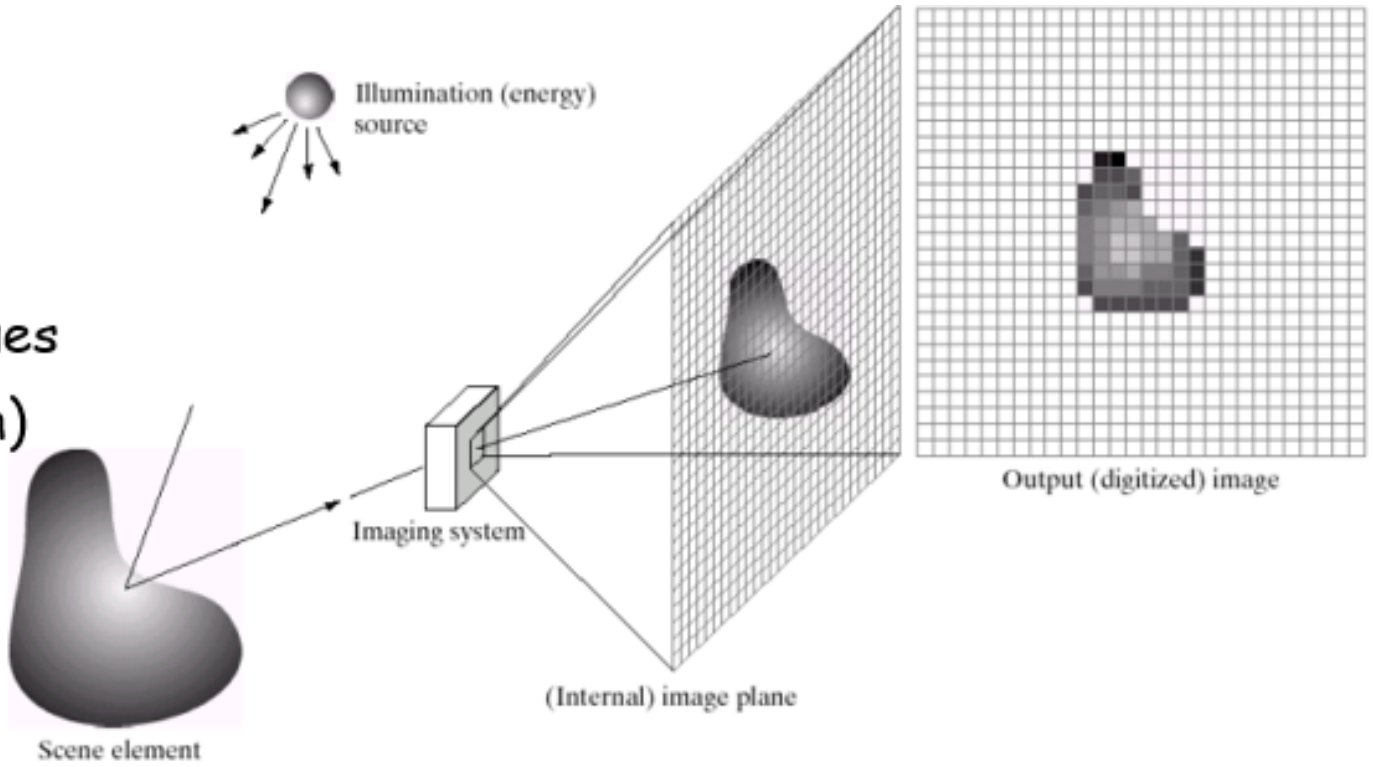
Quantization refers to digitizing the amplitudes, while sampling refers to digitizing the coordinate values.

Sampling

digitizing

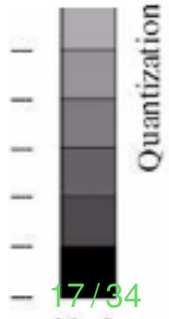
the coordinate values

(Spatial Resolution)



Quantization:

digitizing the amplitude values (Magnitude Resolution)



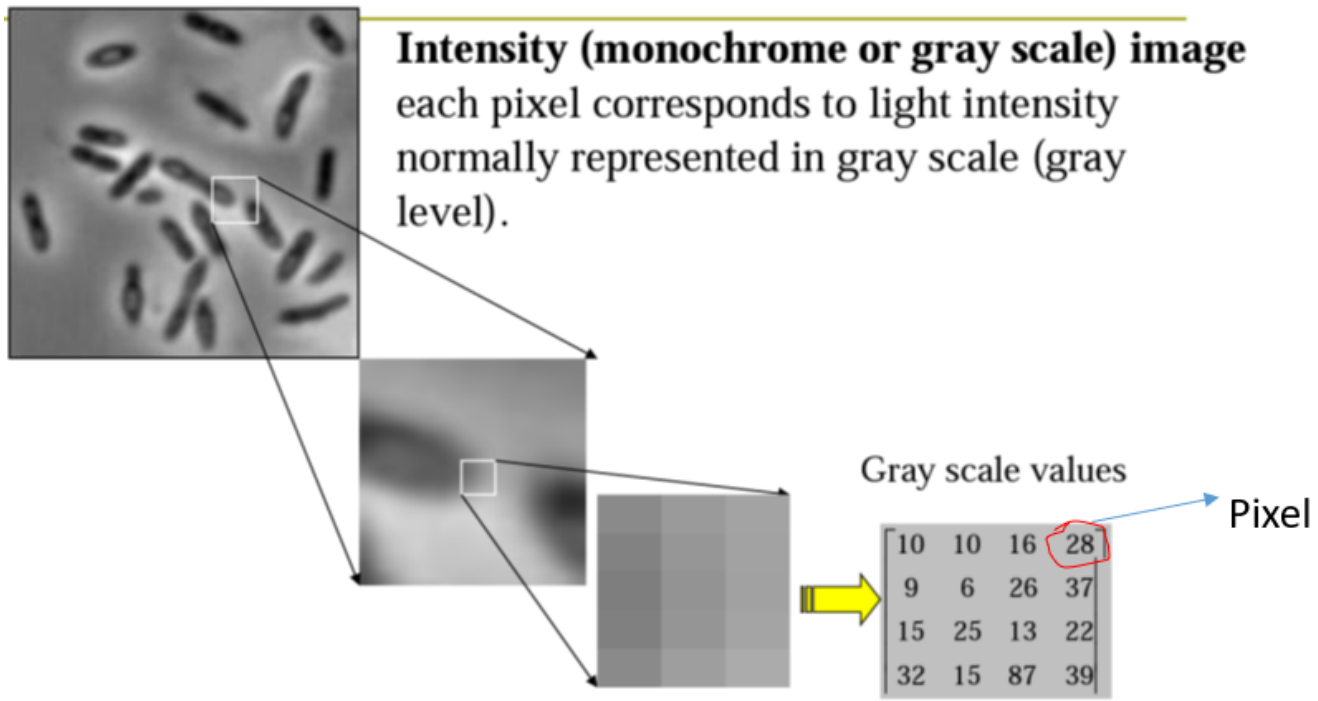
What is Digital Image?

At first, images are considered as gray image.

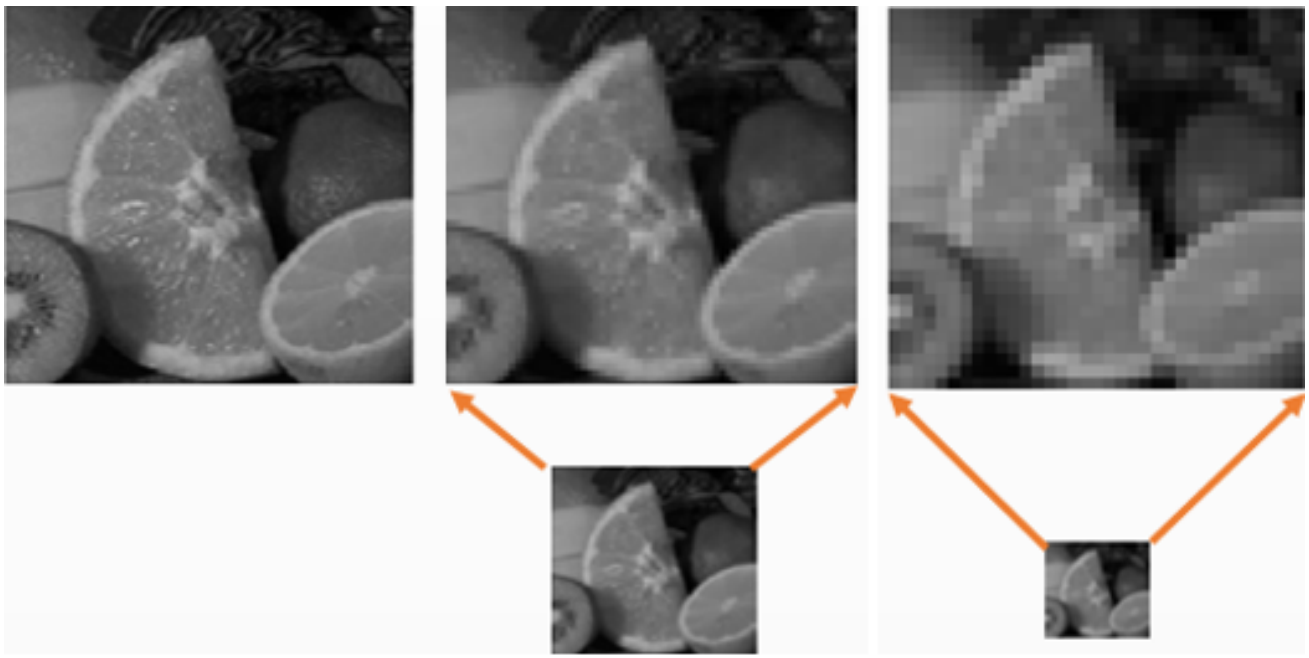
Imgae = A Matrix of Values:

A picture element= Pixel = An element of matrix.

Range Values: $0 - 1$: 256 levels or $0 - 255$: integer 256 levels.



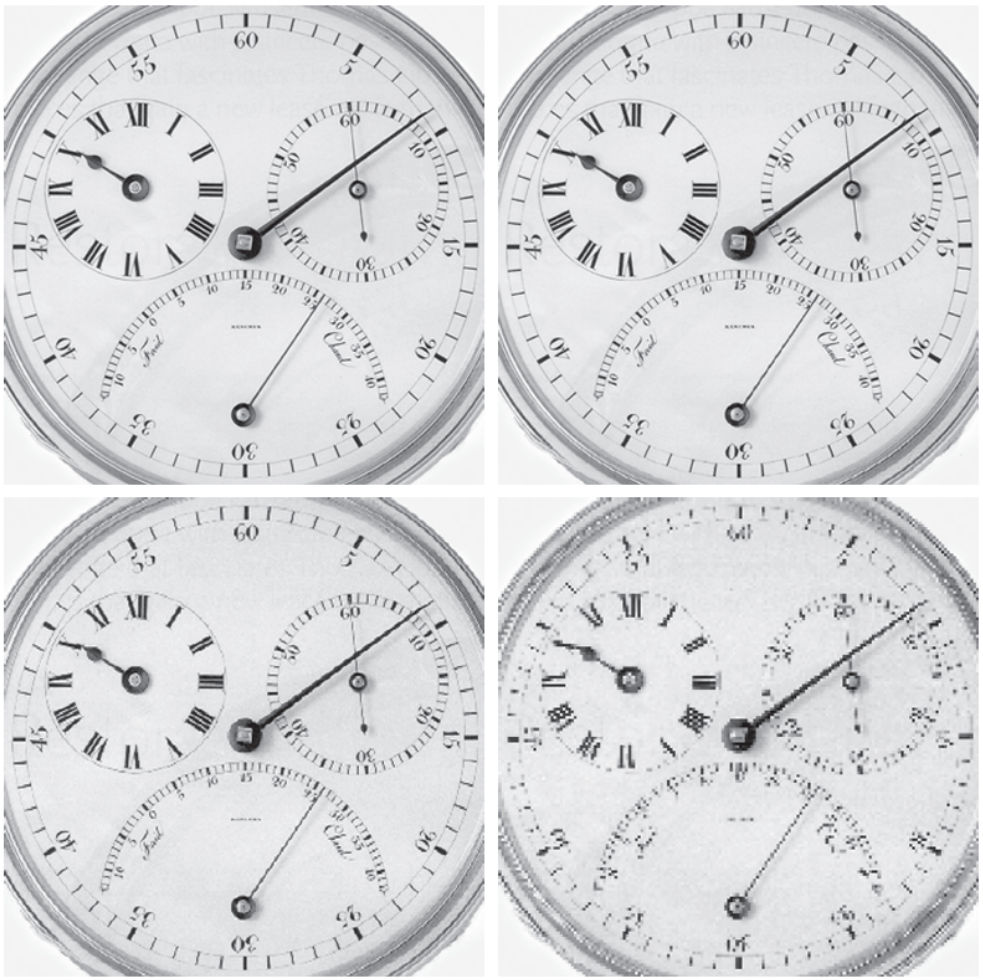
Sampling: EX. 1



Sampling: EX. 2

a b
c d

FIGURE 2.23
Effects of reducing spatial resolution. The images shown are at:
(a) 930 dpi,
(b) 300 dpi,
(c) 150 dpi, and
(d) 72 dpi.



Quantization: How many bits per pixel?



8 bits



5 bits



4 bits



3 bits

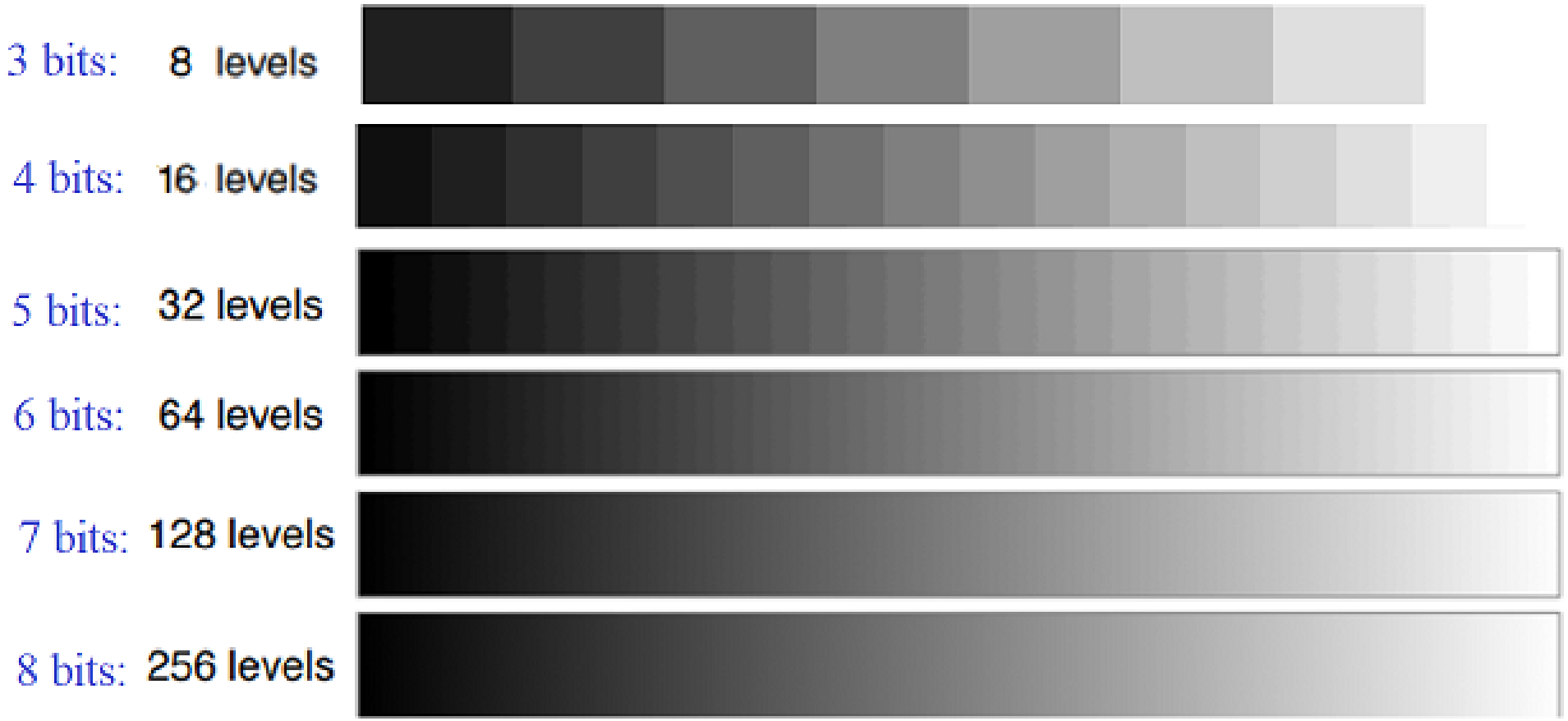


2 bits



1 bit

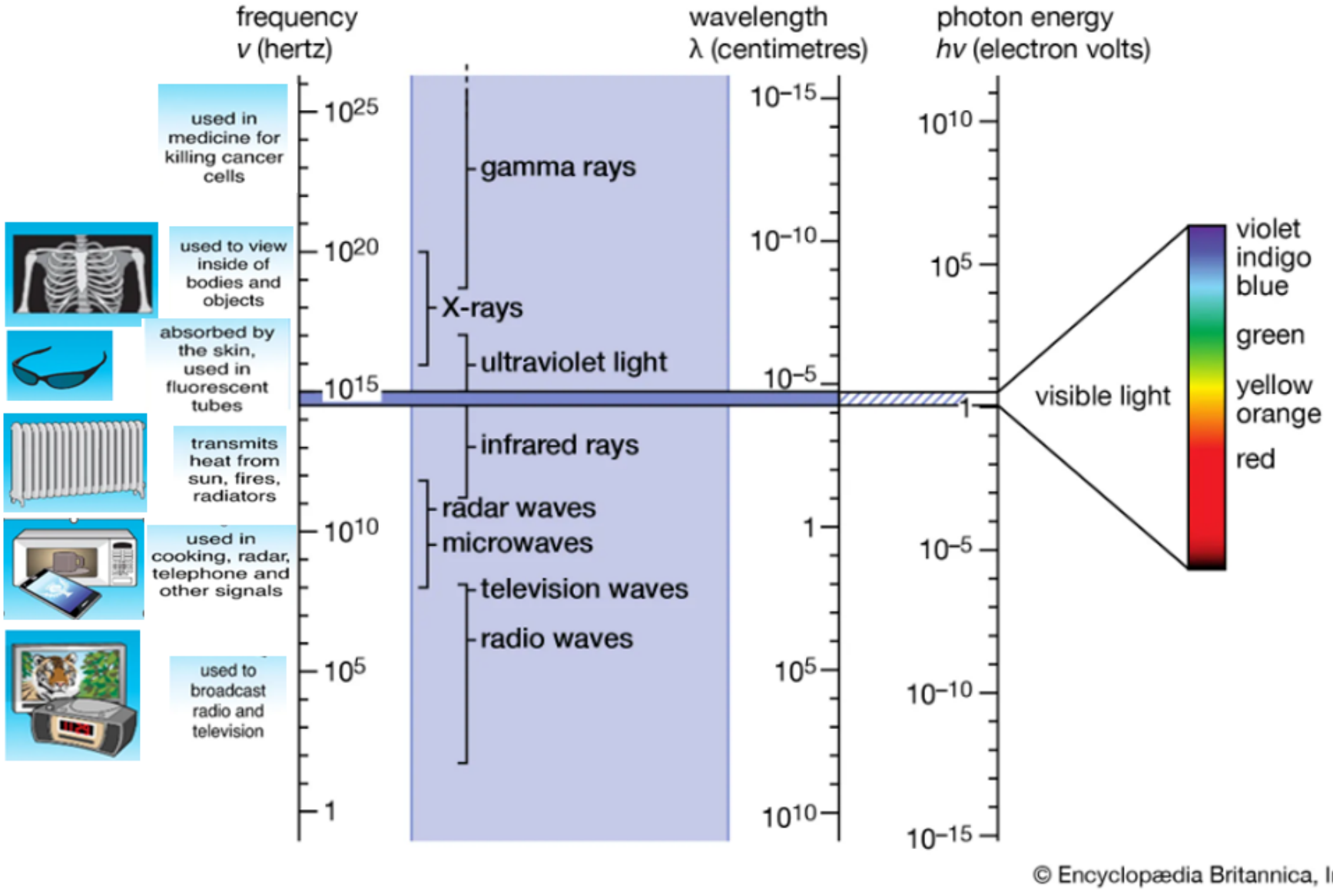
Gray Levels


3 bits: 8 levels
4 bits: 16 levels
5 bits: 32 levels
6 bits: 64 levels
7 bits: 128 levels
8 bits: 256 levels

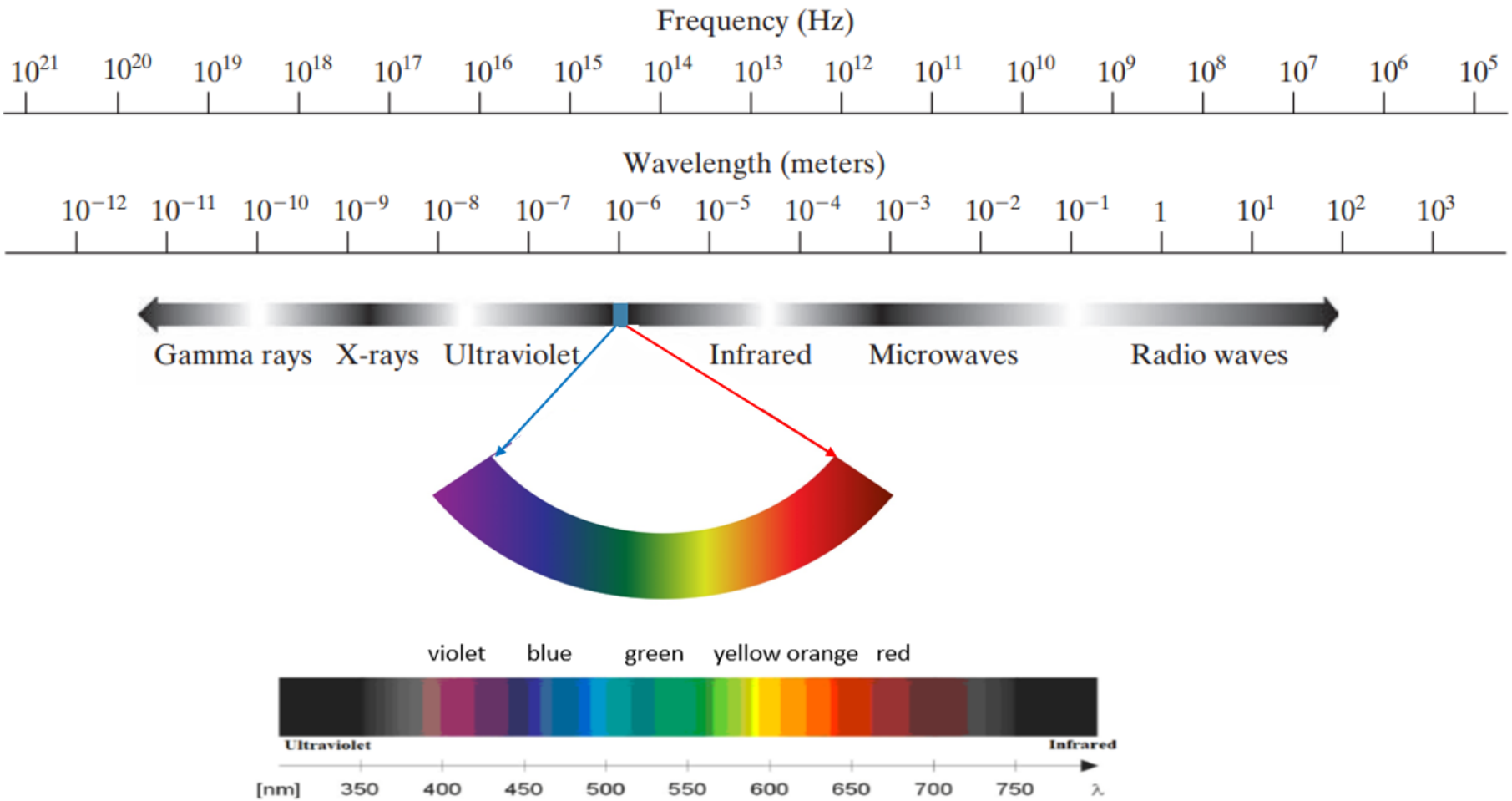
How does human see and process?

To comprehend ***images***, one must first examine how the human brain processes visual stimuli.
The human response to a visual stimulus depends on many factors, including the particular task set by the observer.
At the simplest level, we may be interested in detecting (is it present?) or identifying (what is it?) a stimulus that may be just a small spot of light.
It takes 400-500 milliseconds on average for us to respond to visual stimuli.

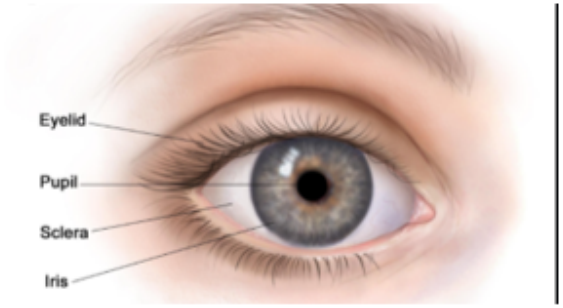
Electromagnetic Waves

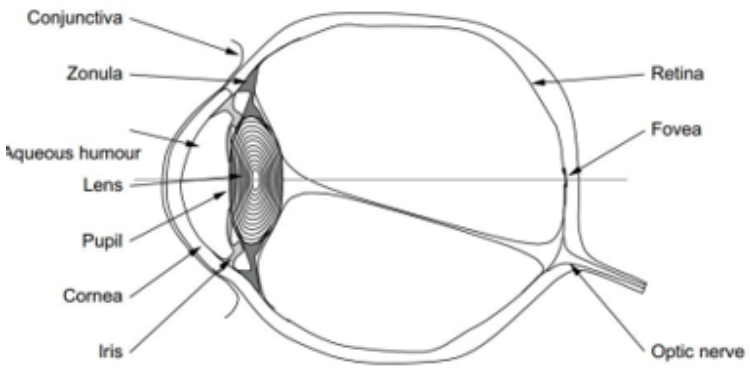


Which waves could be seen?



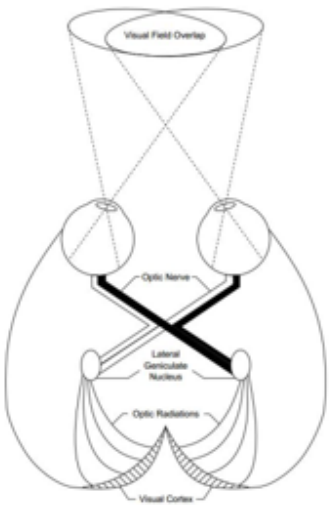
Eye Structure:





Top-down view

The signal path:

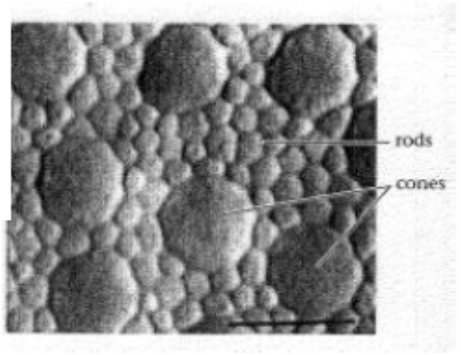


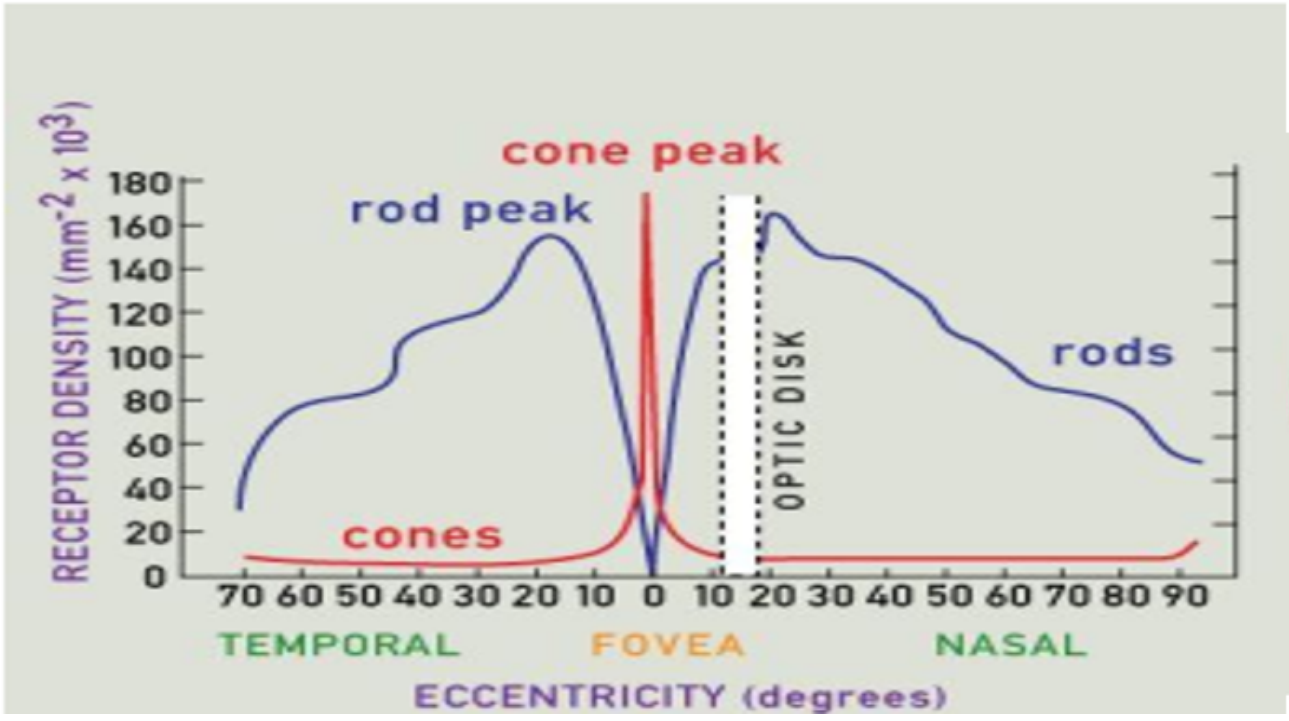
- Lens control: muscle group called *zonula*, changes the shape and position of the lens
- Aperture control: *iris* is a muscle that change the size of pupil.
- Photon sensors: the back of the eye is called *retina*, photo sensor cells concentrate around *fovea*
- Blind spot: where optical nerve terminates

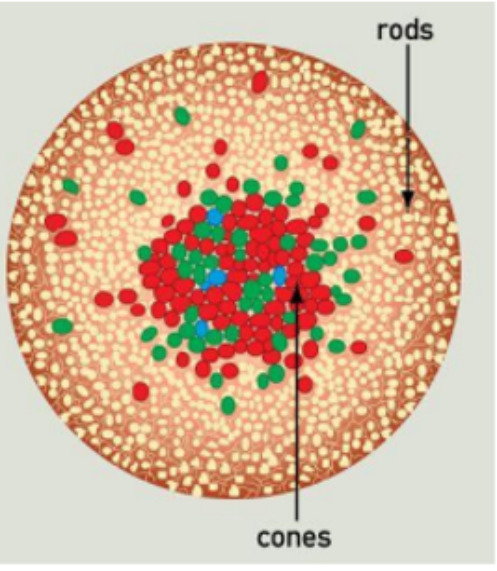
Eye Structure: Continued

Retina photon sensor cells

- Approx. 120 million *rods*
- Approx. 6 million *cones*
- Approx less than 1 million optical nerves (*ganlion*) connecting to brain

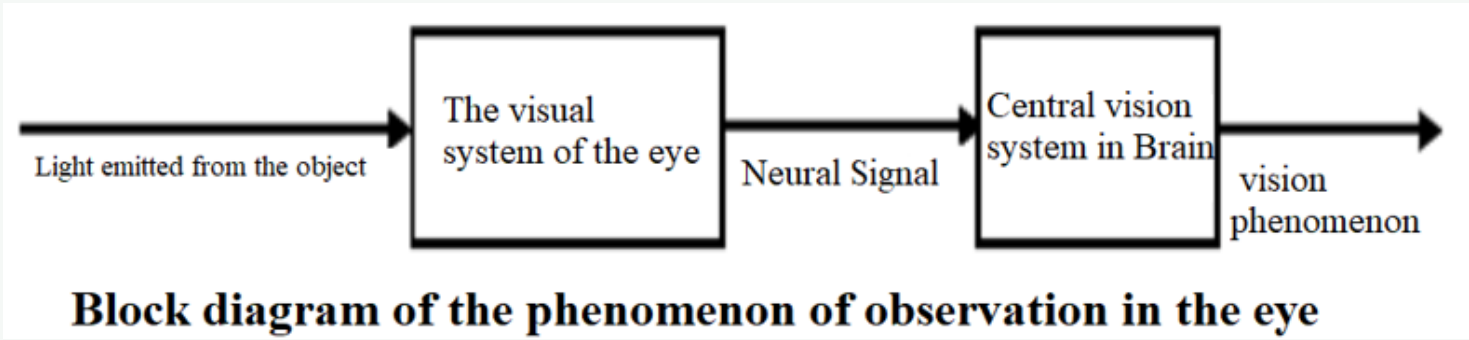






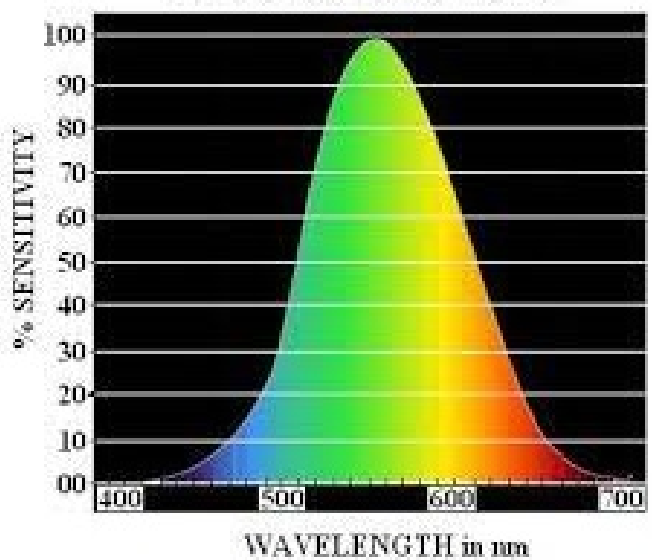
Eye Properties:

The phenomenon of seeing through the eyes

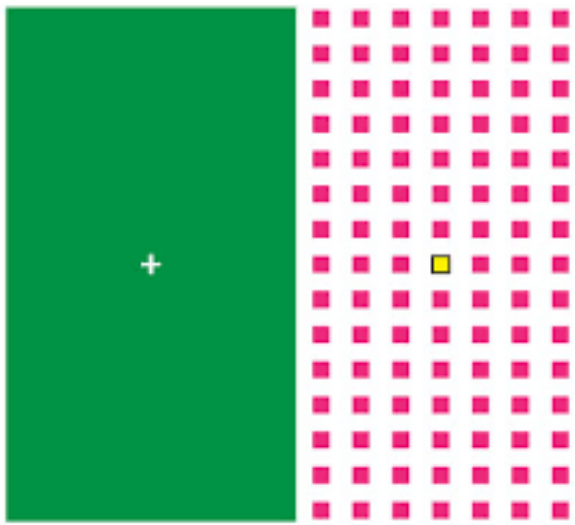


Block diagram of the phenomenon of observation in the eye

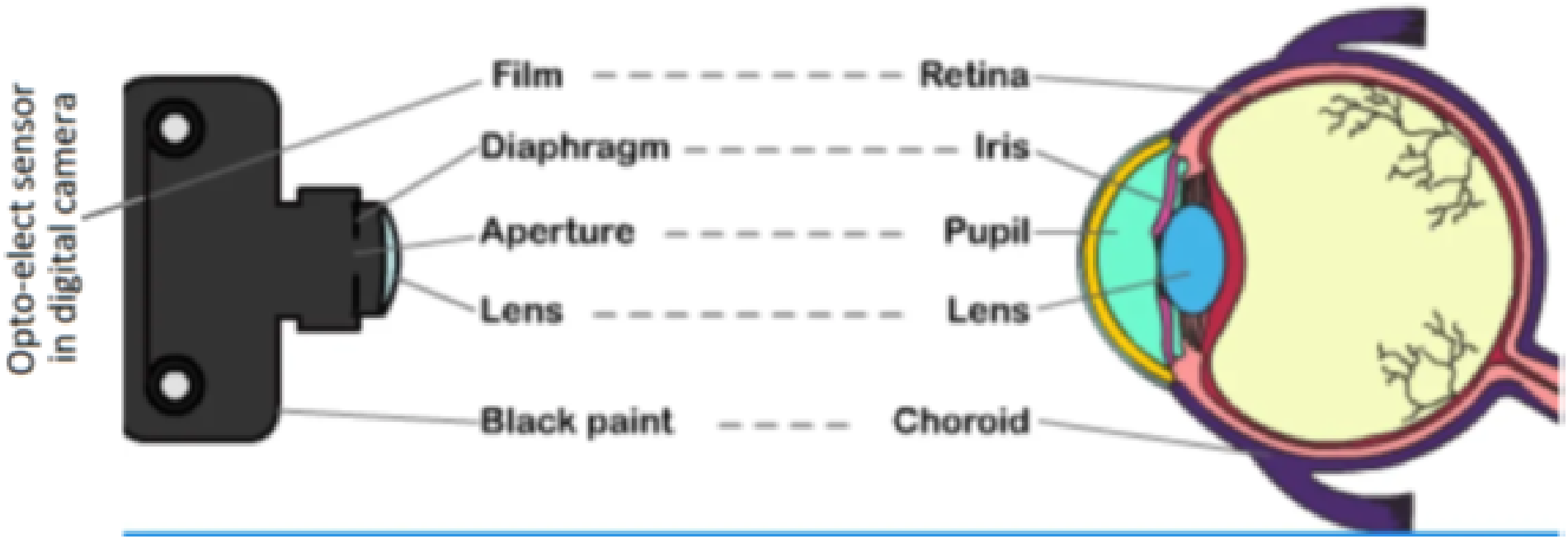
Human Sensitivity



Blind Spot Test

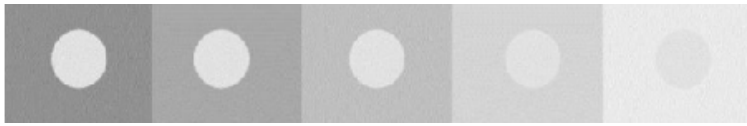


Eye and Camera Similarities



Contrast: Just Noticeable difference (JND)

Circles are in the same brightness.



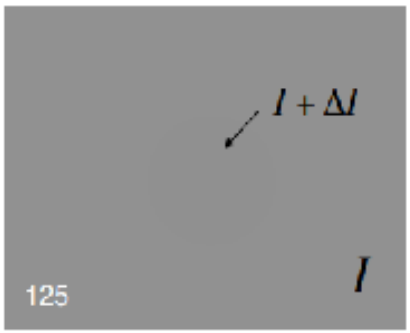
The Just Noticeable Difference (JND) is the difference that allows an observer to detect the center stimulus 50% of the time.

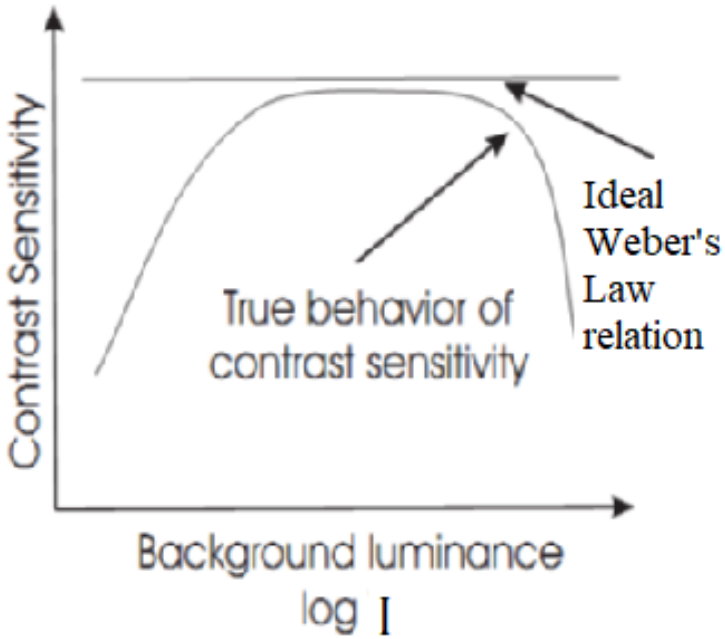
Visibility threshold

$$\Delta I / I \approx 2\%$$

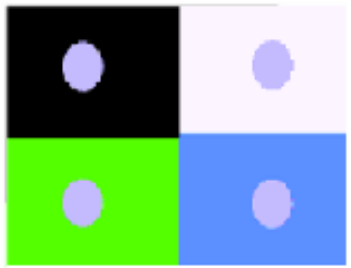


Weber's Law

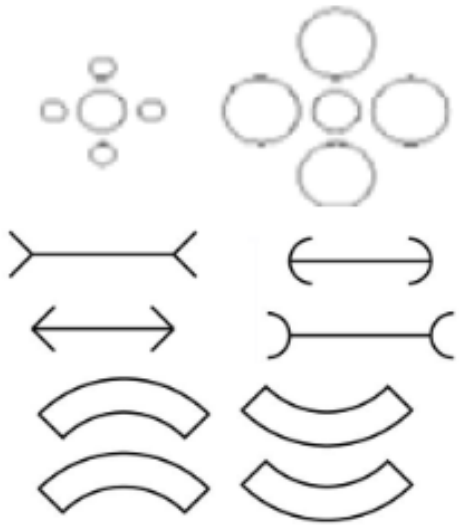




Contrast in colors



Contrast in size



Other Examples of Vision error

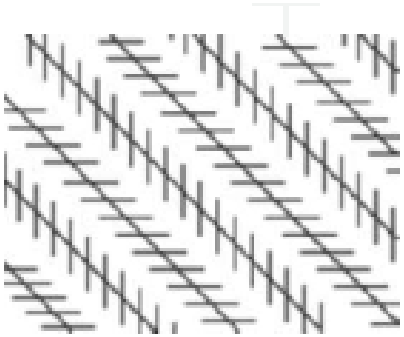


Figure 2.19: Disrupted parallel diagonal lines. © *Cengage Learning 2015.*

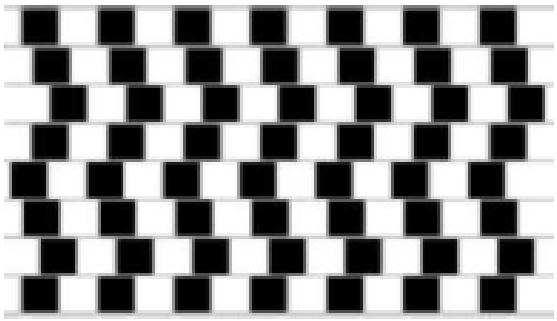
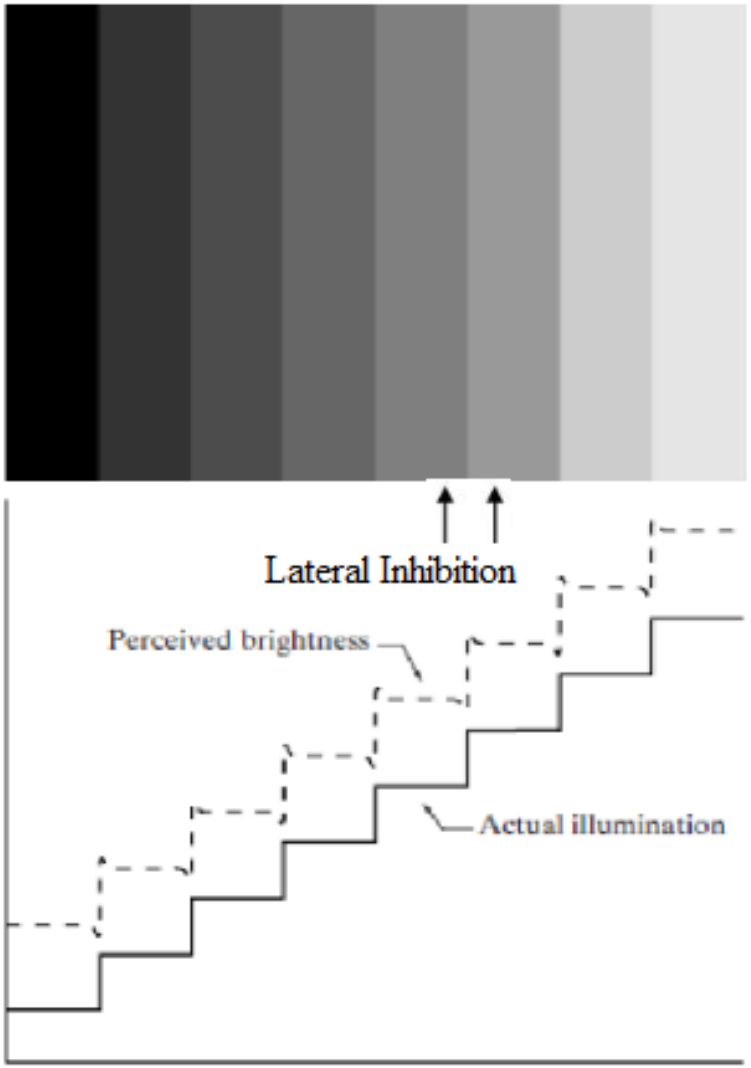
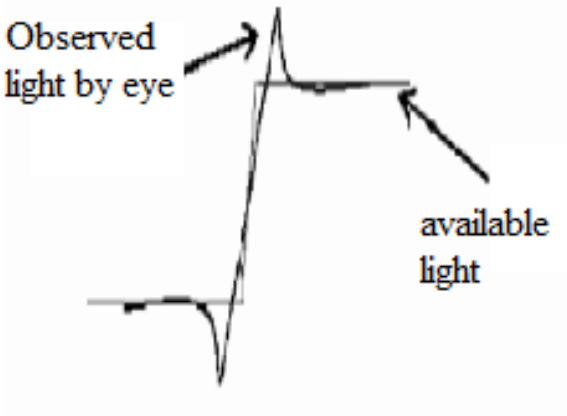
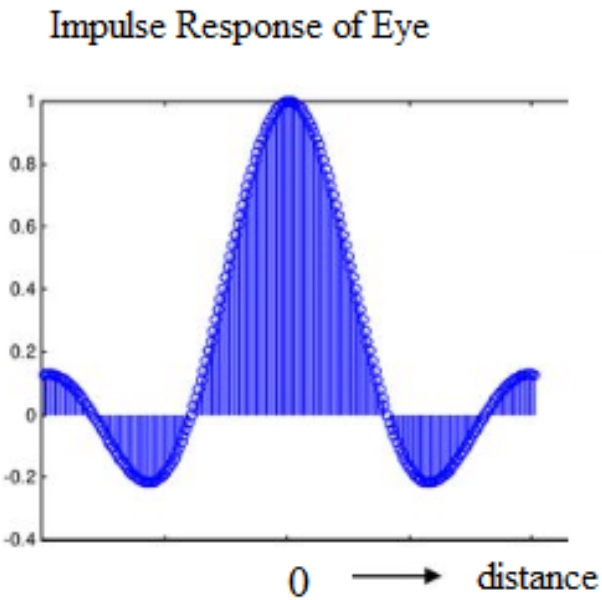


Figure 2.20: Horizontal lines are parallel, although not perceived as such. © *Cengage Learning 2015.*

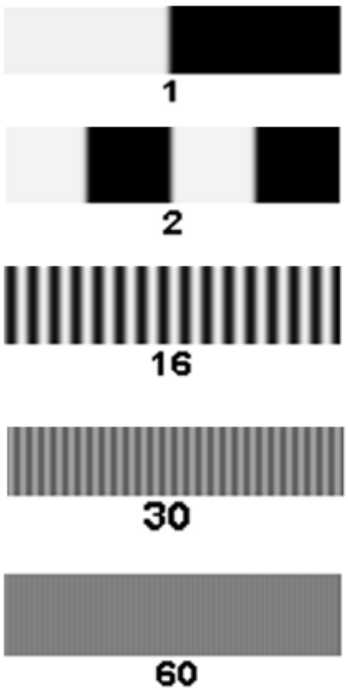
Match Band

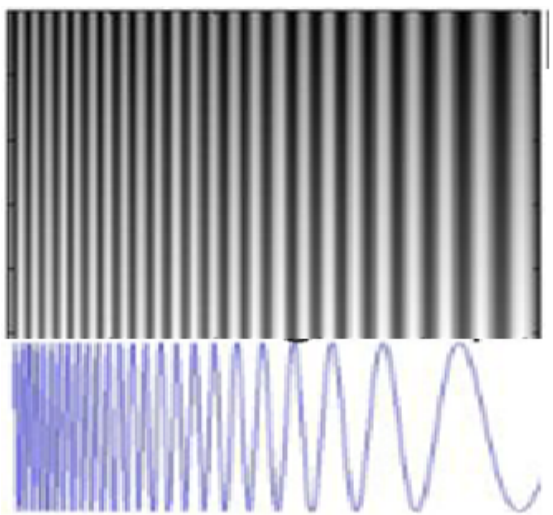

Lateral Inhibition
Perceived brightness
Actual illumination

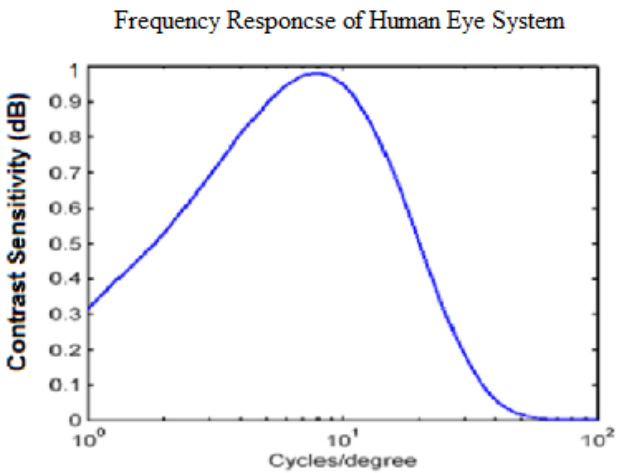

Observed
light by eye
available
light


Impulse Response of Eye
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
0
distance

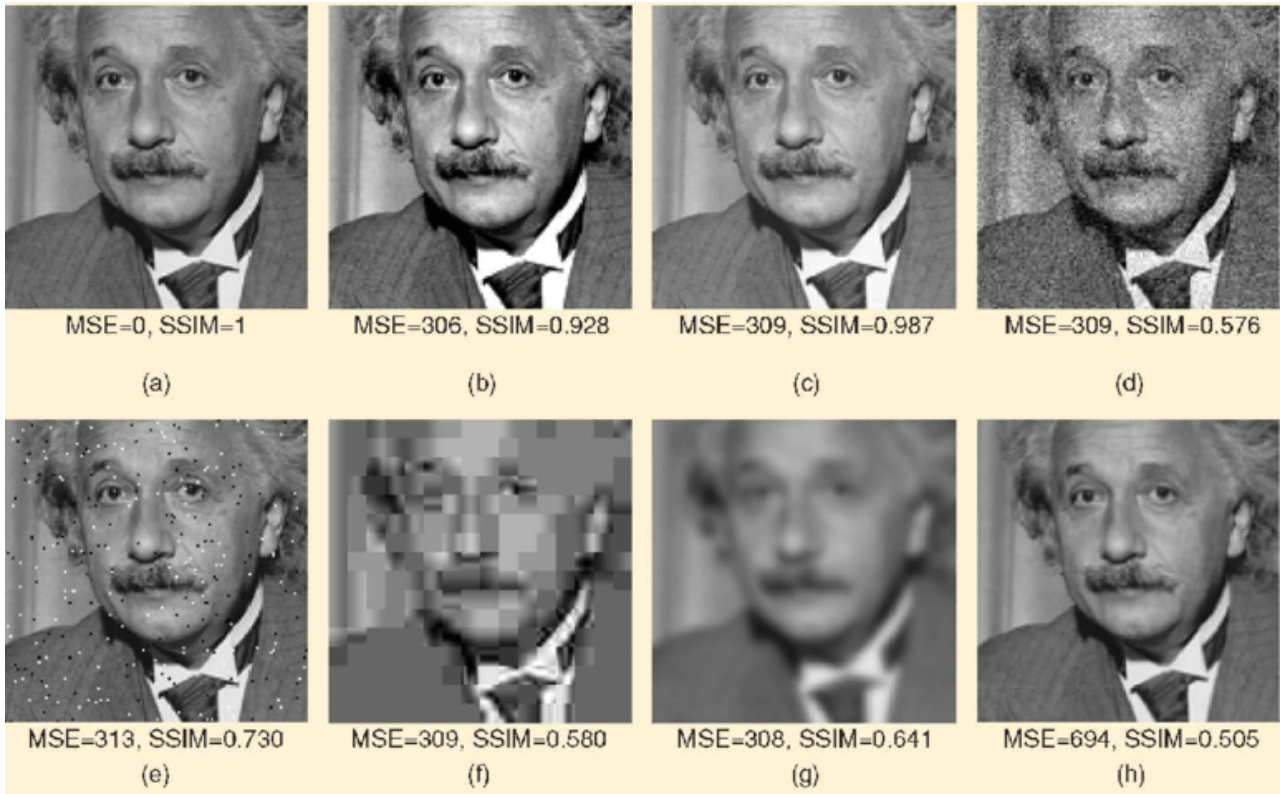
Modulation Transfer Function (MTF) of the Human System







Quality Understanding



If X and Y are the reference image and distorted images, respectively.

$$MSE = \|X - Y\|^2; \qquad SSIM = \frac{(2\mu_X\mu_Y + c_1)(2\sigma_{X,Y} + c_2)}{(\mu_X^2 + \mu_Y^2 + c_1)(\sigma_X^2 + \sigma_Y^2 + c_2)}$$

Yeganeh, Hojatollah, "Cross Dynamic Range And Cross Resolution Objective Image Quality Assessment With Applications" Ph.D. Thesis, Waterloo Uni., 2014.